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PETITEVID Mark 2

A TECHNICAL DESCRIPTION

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GENERAL SUMMARY

The New Bear "Petitevid" is a single-board 'electronic typewriter' video controller based on the Thomson-CSF SFF 96364 i.c. It will cause a suitable TV monitor (composite video input, 625 line, 50 Hz frame) to display 16 rows of 64 upper case ASCII characters, according to the contents of the on-board 1K (6 bit) memory. The memory is loaded sequentially via a blinking (2Hz approximately) 'underscore' cursor, which itself is directable by the input serial start/stop-encoded ASCII character stream.

A UART converts the received serial data to its parallel form and will also convert up to 7 parallel bits of strobed data (from, say, a keyboard) to its serial form. Transmitter and receiver work in the same start/stop code format and at the same baud rate. Serial data line signals are within V24 (RS 232 C) specifications.

Additional features of the Petitevid include:-

- a) "Clear Screen" (and home cursor to top left of screen), from keyboard or via local switch;
- b) Automatic scrolling after sixteenth line;
- c) Home cursor, without clearing screen, from keyboard;
- d) User-strappable positive/negative video;
- e) Three independently adjustable and locally switched UART data rates (up to 120 characters per second--line data rate varies with format chosen; see notes, later, on how to alter "stop" bits, etc.);
- f) By on-board modifications, user may select various serial data formats differing from that supplied--see 'User Options', later;
- g) On-board regulation of "+5v" supply from, say, the now commonly used "+8v" available from micro-computer mainframe.

Power supplies required are:

-12v ($\pm 5\%$) @ 100mA

0v (common ground)

+7v $\rightarrow$ +12v (for "+5v") @ 600mA

CIRCUIT DESCRIPTION

Notes 1) Refer where necessary to main circuit diagram.

2) Inputs and outputs of i.c.s will not be mentioned unless their logic level causes a variation in the circuit's behaviour. Thus, although (say) "chip select" inputs are important in the RAM, their state does not change so they will be ignored. As an exception to this rule, the section on 'User Options' describes how re-wiring of certain UART pins, normally held at a constant logic level, will alter the serial codes generated and decoded by the unit.

3) Positive logic is assumed throughout. That is: low voltage represents "0"; high ($\geq 2.5\text{v}$) voltage represents "1".

4) Most of the i.c. packages contain one functional circuit each. The few which contain four simple gates will be distinguished by reference to their output pins: e.g., "gate P15:3" is the Schmitt NAND gate connected to R3 and C2, whereas "pin P15:3" is the output itself.

UART

The UART (Universal Asynchronous Receiver-Transmitter), if connected to the p.c.b. as supplied, interconverts 7-bit parallel data and serial data of the format shown in Figure 1.

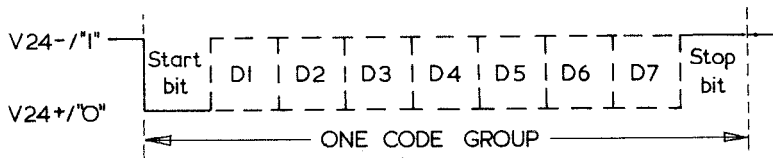


Figure 1: format of Petitevid V24 serial signal (as supplied).

Variations possible (see 'User Options') include: 5-8 data bits; parity bit, odd or even; 1 or 2 'stop' bits; differing transmit and receive baud rates (with an extra clocking input to board).

Note that the same data format is used on Transmit and Receive, and that when the V24 lines are idle they should be at logic "1".

TRANSMITTED DATA

Data is presented on edge connectors (EC) 10-16 (D7-D1, respectively) or on pins 13-26 of the Scotchflex connector. A rising edge on EC17 ("Data Strobe") causes a "0" pulse to appear at pin P6:23 ("DS"); the end of this pulse strobes D1-D7 into the UART, which proceeds to generate a single code group, at pin P6:25 ("SO"--"Serial Output"), embodying these seven bits. The output baud rate is one-sixteenth the frequency of the pulses fed into P6:40 ("TCP") by the oscillator comprising P16, R1, R2, C1. The output signal from P6:25 is inverted and buffered by TR3 with its associated components, appearing at EC35 as "V24 OUT".

RECEIVED DATA

Serial data is received at EC33 ("V24 IN") and passed, buffered and inverted by TRL with its associated components, to pin P6:20 ("SI"--"Serial Input"). The UART detects the onset of a possible 'start' bit as a logic "0" level; but it waits for half a bit period before committing itself. If the "0" level persists for half a bit period--and one bit period is sixteen times the period of the pulses fed into P6:17 ("RCP")--the ensuing waveform at P6:20 is assumed to represent a code group, which is then decoded.

The seven data bits appear in due course at P6:6-12 ("RD7"--"RD1", and D7-D1, respectively); and P6:19 ("DAV"--"Data Available") rises to "1" halfway through the (first) 'stop' bit, about 3µSec after data bits. TR5, along with its associated components, returns a "0" signal to P6:18 ("RDAV"--inverse "Reset Data Available"), approximately 1mSec later, resetting "DAV" to "0". Thus, P6:19 emits a "1" strobe pulse of about 1mSec duration when the seven data bits are available for processing by the rest of the VDU. The seven data bits persist, even after DAV has been reset, until they are over-written by the next seven.

Responses to received data

Of the 128 possible ASCII characters, a number (such as 'BEL') are ignored by the VDU, whilst a few others are reserved for cursor control. The various classes of character are distinguished by P7, a New Bear proprietary ROM. In response to the seven data bits from the UART and the "Clear Screen" external control (of which more will be said later), P7 generates four output bits at P7:9-12. Three of these go directly to the SFF 96364, entering at P8:23-25, and advise it what action is to be taken. The fourth (at P7:12) indicates whether any writing into the on-board display memory is to be permitted. The control characters and their effects are:--

Table 1: *Petitevid control codes.*

ASCII	P7:12	P8:"C2,1,0"	Overall effect (and execution time, mS)
CTRL SHFT "O" VS(0011111)	1	0 0 0	Clear screen and home cursor to top left corner (132).
GS(0011101) } ESC(0011011) }	0	0 0 0	Home cursor to top left corner (132).
CTRL "J" LF(0001010)	0	0 1 0	Line feed: drop cursor by one line (8.3).
CTRL "Y" EM(0011001)	0	1 0 0	Back space: move cursor one character left (8.3).
CTRL "X" CAN(0011000)	0	1 1 1	Forward space: move cursor one character right (8.3).
CTRL "K" VT(0001011)	0	1 1 0	Move cursor up by one line (8.3).
All other CTRL chars + 'DEL'	0	0 1 1	Ignore received character: ie, no action (8.3).
Alphameric	1	1 1 1	Normal character into memory (8.3).

When the seven ASCII data bits become available from the UART, the four control signals are generated by P7. Shortly thereafter (about 3µSec), "DAV" rises to "1", strobing "C0"-"C2" into P8, which proceeds to act on the instructions.

"Clear Screen"

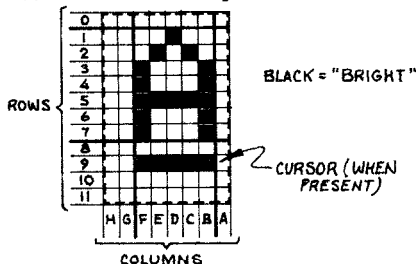
The local "Clear Screen" control forces PROM P7 to generate the output code "1000" (P7:12,11,10,9 respectively) and causes the UART to transmit a 'nonsense' character; assuming the system is connected such that the Petitevid receives all its own transmissions, either by direct wire loop or by computer-echo, "DAV" will go to "1" shortly thereafter and cause the SFF 96364 to act on the locally-forced code.

Video Signal Generation

It is convenient at this point to digress and discuss how the proper output video signal is generated, drawing on the RAM's contents.

A symbol is built up by the TV raster by elevating the video signal from 'dark' to 'bright' level whenever the raster crosses part of it. For the sake of simplicity, all Petitevid symbols are formed from points on a dot matrix, 5 units wide and 7 units high. Furthermore, to allow a margin around the symbols, they are separated by being nested within a larger matrix, 8 units wide and 12 units high (although some TV monitors will cause the non-interlaced video signal to be 'doubled', so displaying twice as many units vertically). Figure 2 shows an upper case "A" as an example. Note that the symbol area extends from Row 1 to Row 7 and from Column F to Column B.

Figure 2: construction of a 'Petitevid' screen symbol.

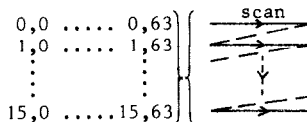


also that the cursor, when present, lies on Row 9, Columns F-B. Symbol and cursor do not, however, appear on the screen simultaneously, but alternately, whilst symbols elsewhere are displayed continuously.

Memory Read-Out

(Refer to Figure 3.) The 1024 locations within the RAM are organised as successive screen characters, numbered from Line 0, Char.0, through to Line 15, Char.63.

Figure 3: relationship of memory locations to screen character position.



As the TV raster moves down the screen, it makes twelve scans through Line 0 (RAM locations 0-63), then twelve through Line 1 (RAM locations 64-127), and so-on. On each scan, as it accesses a character, the 6-bit outputs pass from P0-P5 to P9, where the ASCII code is staticised for one screen character period. The outputs of P9 then pass to P10, the character dot matrix ROM; there they are combined with the information (from the SFF 96364) as to which row of the character is being generated and whether or not the cursor is present. (P8:15 is "1" if the cursor is present; and P8:11,12,13 depend on the row number in a moderately involved way.

The values which appear at RO₂-RO₀ for each row are shown in Table 2.

Table 2: relationship of character row number to values of RO₂-RO₀ (P8:13-11 respectively).

Row No.	RO ₂	RO ₁	RO ₀
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0
9	0	0	0
10	0	0	0
11	0	0	0

The ROM, P10, is programmed to generate 'all-0' outputs whenever P10:14-16 are "000". Thus rows 0,9,10,11 are always 'dark'. The cursor input, which appears in row 9, over-rides normal operation by making the outputs 'all-1', except during other rows, which are 'all-0'; this, of course, applies only whilst the blinking cycle is in its 'on' phase, and is achieved by RO₂-RO₀ being "000" during non-cursor rows.

Using the character and row information, P10 defines which pattern of 'bright' ("1") and 'dark' ("0") levels is to be impressed on the video signal during the next character scan period. This it does by passing the five character row elements (F-B) to the 8-bit shift register, P11, into which they are loaded along with columns H,G and A; these last three, being 'dark' for all symbols, are tied to "0".

During the next character scan period, even whilst information for the next two characters is being prepared by P0-5, P9 and P10, eight rapid clock pulses shift the character elements out of P11:9 (or P11:7 for inverted video), through the TR4 buffer amplifier to the output socket.

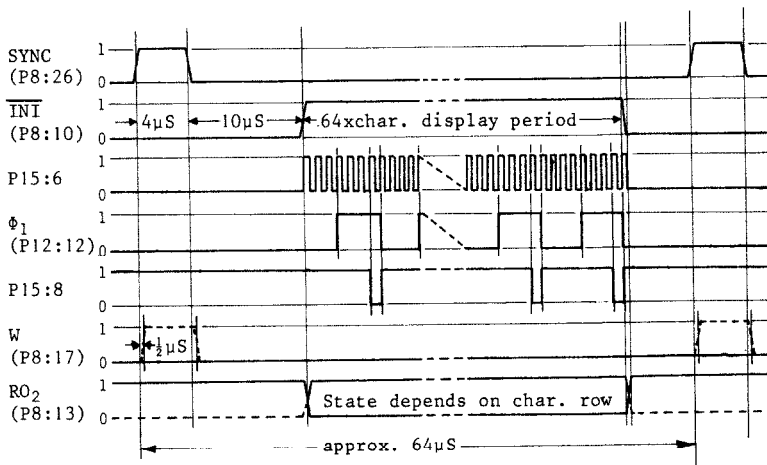
The clocking signals for each stage of this process are generated by a combination of the SFF 96364 interworking with the following i.c.s:

- P15:3 (character element shift-out rate oscillator);
- P15:6 (inverting buffer to sharpen edges of clock waveform);
- P12 (divide-by-8 counter);
- P15:8 (inverter and buffer for character-rate clock).

The SFF 96364 times all its synchronisation signal and line scan processes from its own internal crystal-controlled oscillator. P8:2 and P8:1 are, respectively, the output to, and input from, the 1008kHz crystal.

The SYNC (P8:26) output during a normal line period is shown in Figure 4, along with the clock waveforms at P8:10 ("INI"), P15:6, P12:12 (" Φ_1 ") and P15:8. These clock waveforms are included to clarify the foregoing explanation; the bottom two waveforms, those at P8:17 ("W") and P8:13 ("RO₂"), are relevant to the explanation of the RAM write-in process which follows.

Figure 4: some important waveforms relating to RAM read-out and write-in processes. (Idealised.)



Memory Write-In

RAM contents can only be altered if P14:11 goes to "0". Reference to Figure 4 shows that signal "W" (P8:17), which is gated with the controlling PROM signal from P7:12, will only permit this condition during the 'SYNC high' period, during line resynchronisation when no character information is being imposed on the video waveform. During this period, if the input control code at "C0"- "C2" (P8:23,24,25) specifies any kind of 'write' operation, the cursor address (its location on the screen) is put onto the "RAM ADDRESS" lines, so that the appropriate RAM location will be altered.

Two main types of 'write' operation can occur:

1) A single ASCII character. In this case, RO₂(P8:13) remains at "1" throughout the "INI=0" period, permitting data to pass from the UART, through the cluster of seven gates, to the RAM inputs. The RAM is loaded with slightly modified 'standard' ASCII; the modification, a rearrangement of the most significant bits, causes an automatic conversion of lower to upper case.

The full ASCII character set may be divided into four groups, each defined by its two most significant bits:--

	Bits 7 6 5 4 3 2 1	
Control	0 0 x x x x x	('x'="0" or "1", as required.)
Figures, etc	0 1 x x x x x	
U/C letters	1 0 x x x x x	
L/C letters	1 1 x x x x x	

For 'upper case only' operation, bit 6 may be dispensed with, since PROM P7 automatically prevents control characters from being recorded. This, it will be seen, is what is done in the Petitevid. Furthermore, bit 7 is inverted, and effectively is transformed into a 'replacement' bit 6.

2) Multiple erasures. In this case, RO₂ goes to "0" during the "INI=0" period. The effect is to force the RAM data inputs to "100000", which is the abbreviated-ASCII code for "space". Then, depending on the code specified at "C0"- "C2", the SFF 96364 carries out multiple 'write' operations, moving the cursor as required between each, until the entire task specified has been completed.

Video Output

In conclusion, the symbol element information from P11 is combined with synchronisation signals from the SFF 96364 in a weighted summation circuit (R5 and R6) and passed thence, through the TR4 emitter-follower buffer, to the co-axial output socket, SK2.

USER OPTIONS

Extensive though the Petitevid's capabilities are, it does not exploit to the full the potential inherent in the UART or the SFF 96364. To do so would not be appropriate to a 'basic' VDU; users who wish to explore further are referred to the data sheets for these devices and are cautioned that New Bear regretfully cannot enter into technical discussions of developments not described here.

However, some useful options are available to users by means of simple board modifications. They will be listed briefly below:--

1) Text Width: increasing the time constant R3.C2 will extend the rows of characters further towards the right hand side of the screen. The starting point is unaffected.

2) Video Modulation Sign:

Positive: strap LK1 from R6 to P11:9, LK2 from R5 to P15:11.

Negative: ----- P11:7, ----- P8:26.

3) Baud Rate Setting: if R1 (A, B or C) proves hard to set accurately, fixed parallel or series resistors should help, provided values are chosen carefully to reduce effect of the variable resistors' 20-turn range limit. C1 may also be altered.

4) UART options: a number of UART pins, if reconnected to 0v ("0") or +5v ("1"), will alter the V24 signalling from that supplied. (NB: remove UART from its socket before attempting any work on the p.c.b.) The fullest form of the serial data is:--

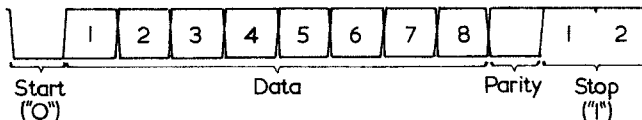


Table 3: UART options connection list.

FUNCTION	VALUE	PIN n = 0/1	COMMENTS
Parity bit	None	35(NP)=1 39(EPS)=0/1	Pin 13 ("Parity Error") goes to "1" if the received character parity does not agree with the selected parity.
	Even	35=0 39=1	
	Odd	35=0 39=0	
Stop bits	One	36(TSB)=0	Pin 14 ("Framing Error") goes to "1" if the received character has no valid 'stop' bit.
	Two	36=1	
No. of data bits	5	37(NB2)=0 38(NB1)=0	
	6	37=0 38=1	
	7	37=1 38=0	
	8	37=1 38=1	

5) Different Transmit and Receive Baud Rates: in principle, the UART's outgoing and incoming baud rates may be made different if either pin 40 (tx) or 17 (rx) is split off and driven by a separate TTL-level oscillator at 16x the required rate.

4

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1.